Capstone Report:

AI Governance Framework: Equitable Pediatric Diabetes Diagnosis (Canada)

April 30, 2026

1.0 Purpose, Scope & Guardrails

As Alberta Health Services (AHS) continues to adopt clinical AI tools to support early and accurate diagnosis, a critical gap persists in ensuring that these models perform equitably across diverse patient populations once deployed in real-world settings. In the absence of a standardized and repeatable governance approach, there is a risk that models may unintentionally underperform certain demographic, geographic, or socioeconomic groups—potentially leading to undetected disparities in care.

The Gatekeeper dashboard is designed to address this gap by providing a structured, system-level view of model performance and diagnostic equity. It enables the proactive identification of bias and variation across patient subgroups before these differences translate into adverse clinical outcomes.

For the purposes of this pilot, the system is applied within a synthetic pediatric diabetes context. The data used is artificially generated to reflect realistic patient distributions and clinical patterns, while ensuring that no real patient information is included. The scope is intentionally constrained to pediatric patients under the age of 18 and to diabetes-related diagnostic support models, including Type 1 and Type 2 diabetes where applicable. Adult populations, gestational diabetes, and unrelated endocrine conditions are excluded. As such, findings should not be generalized beyond this defined context without further validation.

The primary purpose of the dashboard is to support AHS data scientists, clinical leaders, and governance stakeholders in making informed and transparent decisions regarding model deployment and monitoring, with a particular emphasis on ethical inclusion across patient populations. By explicitly surfacing how model performance varies across subgroups, the system promotes accountability and helps ensure that vulnerable or historically underserved populations are not overlooked during model evaluation.

The outputs generated by the system are strictly informative. They combine observed results with projected trends, where appropriate, and are intended to guide quality improvement, audit, and research activities, rather than clinical decision-making. The Gatekeeper system functions as a post-model governance evaluation layer, assessing models after development to determine whether they meet predefined fairness and performance thresholds.

Importantly, the system is non-diagnostic and non-prescriptive. It does not generate diagnoses, clinical predictions, or treatment recommendations, nor does it alter or override model outputs in real time. Instead, it produces standardized governance outcomes “Pass,” “Needs Review,” or “Fail” to support transparent discussions on whether a model should be deployed, monitored more closely, or reassessed.

By making subgroup performance both visible and interpretable, the Gatekeeper dashboard reinforces a commitment to equitable care and responsible AI use. At the same time, it maintains a clear boundary as a governance safeguard within the AI lifecycle, rather than a tool for patient-level decision-making.

2.0 Data Governance & Stewardship

Building on this governance foundation, effective data governance and stewardship are essential to ensuring that the Gatekeeper system produces reliable, equitable, and accountable insights. While the broader system supports Type 1 and Type 2 diabetes diagnostic models, this pilot focuses specifically on predicting diabetic ketoacidosis (DKA) risk within 90 days as a representative evaluation use case. While the dashboard evaluates model performance at a system level, the integrity and representativeness of the underlying data directly determine the validity of its outputs.

- **Scope and Data Architecture:** The pilot evaluates pediatric diabetes predictions—specifically the risk of diabetic ketoacidosis (DKA) within 90 days—across the five AHS health zones (Calgary, Edmonton, North, South, and Central). The dataset is intentionally designed to reflect real-world complexity.
    - Data is sourced from multiple EHR systems (EHR_A and EHR_B), introducing variability in structure and completeness.
    - Measurements include both CGM (Continuous Glucose Monitor)-derived and lab-derived glucose values, reflecting differences in clinical workflows and patient access to technology.
    - This multi-system design improves realism but introduces data provenance considerations, including potential inconsistencies, coverage gaps, and variation in data capture practices.
- **Representativeness and Coverage Risks:** Variability in data availability across regions and populations introduces important limitations that must be explicitly acknowledged when interpreting outputs.
    - Rural zones and smaller demographic groups (e.g., Indigenous patients, newcomers, non-English/French speakers) are expected to have fewer encounters.
    - Smaller sample sizes lead to less stable subgroup-level metrics and increased variability in performance estimates.
    - If the dataset skews toward urban, higher socioeconomic status (SES), or CGM-equipped populations, model performance may not generalize reliably to rural, lower-SES, or lab-only populations.
    - These representativeness gaps create a risk that observed fairness metrics may not reflect true real-world performance across all populations.
- **Data Quality and Missingness:** To support transparent interpretation, the dataset includes explicit indicators of data quality, acknowledging that missing data is not random.
    - Each record includes measures such as *missingness_rate_row* and *is_imputed_any* to track completeness and imputation.

- Certain populations—particularly rural patients, lower-SES groups, and those without consistent primary care access—are more likely to have incomplete laboratory data (e.g., creatinine, eGFR, LDL, triglycerides).
    - As a result, the governance system must flag or restrict interpretation of fairness metrics for subgroups with high missingness, where results may be unreliable or biased.
- **Ethical Use of Sensitive Attributes:** The inclusion of sensitive attributes, such as Indigenous identity, requires careful and intentional governance.
    - While the current dataset is fully synthetic and does not pose privacy risks, real-world implementation would require strict ethical oversight.
    - Any use of Indigenous data must align with OCAP® principles (Ownership, Control, Access, Possession).
    - Engagement with First Nations, Métis, and Inuit governance bodies and health leaders is essential to ensure culturally appropriate and respectful data use.
- **Privacy and Access Controls:** Strong safeguards are required to prevent re-identification, particularly when multiple attributes are combined.
    - Risk increases when variables such as age band, health zone, race/ethnicity, and Indigenous identity intersect—especially in small or rural populations.
    - Standard practices should include suppressing subgroup-level results below a minimum threshold (e.g., fewer than 20 patients).
    - Role-based access controls should be enforced, allowing stakeholders to view aggregated insights while restricting access to individual-level data.
- **Governance, Auditability, and Traceability:** To maintain accountability, all governance actions and outputs must be transparent and reproducible.
    - Overrides of system decisions should be logged and restricted to authorized personnel.
    - Every evaluation output must be traceable to the specific policy rules and threshold versions applied.
    - This ensures reproducibility for audit, compliance, and continuous improvement processes.
- **Proactive Equity Safeguards:** As a foundational control, governance must extend upstream into data preparation and model development cycles.
    - A Data Equity Risk Assessment should be conducted prior to any model training or retraining.
    - This assessment should evaluate subgroup representation, missingness patterns, and label distribution differences.
    - The objective is to identify and mitigate risks early, ensuring that models do not reinforce or amplify existing health disparities.

3.0 System Architecture and Operational Workflow

To support the evaluation of model performance, fairness, and data quality described in this report, a structured monitoring system, referred to as the Gatekeeper system, is implemented. This system enables continuous assessment of AI model behavior in real-world clinical environments.

The architecture is designed as a post-deployment monitoring and governance layer that operates independently from clinical workflows. It does not generate predictions or influence patient care. Instead, it evaluates model outputs and translates them into system-level insights to support oversight and decision-making.

At a high level, the system connects clinical data sources, data pipelines, analytics processes, and governance functions into a continuous monitoring workflow.

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Clinical Systems (Hospital Information System – Epic)
        |
        ▼
Secure Data Pipelines
(ETL / APIs / controlled data extraction)
        |
        ▼
Enterprise Analytics Environment
(Centralized analytics platform for monitoring data)
        |
        ▼
Monitoring Computation Layer
(Python + Evidently AI)
        |
        | Computes:
        | • fairness metrics
        | • performance indicators
        | • data and prediction drift
        ▼
Governance Decision Layer
(Pass / Needs Review / Fail)
        |
        ▼
Governance Outputs and Reporting
(Dashboard, reports, governance review)
```

The system operates as an end-to-end pipeline:

- **Clinical systems (Hospital Information System — HIS i.e. EPIC)**
    - Patient data is generated during routine care and recorded within the clinical system
    - Embedded AI models produce predictions (e.g., risk of DKA)
    - These predictions form the primary inputs for monitoring
- **Secure data pipelines**
    - Relevant data, including predictions and clinical information, is extracted through governed pipelines
    - Data movement is controlled to ensure privacy, security, and minimal exposure
- **Enterprise analytics environment**
    - Extracted data is centralized and standardized for analysis
    - Historical data is maintained to enable comparison over time
    - This environment supports reproducible and scalable computation
- **Monitoring computation layer**

    - Model performance, fairness, and drift metrics are calculated
    - These computations transform individual predictions into aggregated insights across patient populations
- **Dashboard and reporting layer**
    - Results are presented through dashboards and structured reports
    - Outputs include subgroup comparisons, trends, and alert indicators
    - This layer enables stakeholders to interpret model behavior at a system level
- **Governance and decision layer**
    - Monitoring outputs are evaluated against predefined thresholds
    - Standardized outcomes (e.g., Pass, Needs Review, Fail) support consistent decision-making
    - Actions may include continued monitoring, investigation, or model reassessment

In essence, the system operates as a continuous cycle:

- Model predictions are generated within clinical systems
- Clinical outcomes are recorded during follow-up care
- Predictions and outcomes are extracted through data pipelines
- Data is processed and analyzed within the analytics environment
- Metrics are computed and aggregated across subgroups
- Results are visualized in dashboards
- Governance decisions are applied based on observed performance

This creates a closed-loop system, where real-world outcomes continuously inform evaluation and monitoring. It also supports reliable and scalable monitoring in the healthcare context to ensure:

- **Separation from clinical workflows**
    - Monitoring operates independently to avoid impacting patient care
- **Continuous evaluation**
    - Model performance and data quality are assessed on an ongoing basis
- **Data governance alignment**
    - Data handling follows privacy, security, and access control requirements
- **Reproducibility and auditability**
    - All computations are performed in controlled environments, enabling validation and audit
- **Scalability**
    - The architecture supports monitoring across multiple models and populations

4.0 Equity Definition & Measurement Governance

Following the system architecture described above, model predictions and clinical data are processed within the analytics environment to evaluate performance, fairness, and stability. This section describes how these outputs are transformed into measurable indicators that support monitoring and governance.

4.1 Model Outputs and Evaluation Inputs

Within the clinical system, the model generates predictions for individual patients in the form of risk scores and classification outcomes (high risk vs low risk). These predictions are recorded as part of routine care. Clinical outcomes, such as confirmed diagnoses or adverse events, are captured over time and provide the ground truth required for evaluating model performance.

Predictions and outcomes are extracted through secure data pipelines and integrated within the analytics environment, where they are matched using patient identifiers and timestamps. This pairing enables each case to be evaluated and forms the foundation for all performance and fairness metrics.

4.2 Interpreting Model Outcomes

Once predictions and outcomes are aligned, each case is categorized into one of four outcome types:

Outcome Type	Definition	Clinical Interpretation
True Positive (TP)	Correctly identifies a patient who develops the condition	Enables timely intervention
False Negative (FN)	Misses a patient who develops the condition	**Critical safety risk (missed diagnosis)**
False Positive (FP)	Incorrectly flags a patient as high risk	Leads to unnecessary follow-up
True Negative (TN)	Correctly identifies a low-risk patient	Avoids unnecessary care

These outcome classifications are generated within the analytics environment and serve as the building blocks for performance metrics.

4.3 Illustrative Example

To illustrate how these outcomes translate into evaluation metrics:

- Out of 100 patients, 20 develop diabetes-related complications
- The model correctly identifies 17 of these patients (**True Positives**)
- The model misses 3 patients (**False Negatives**)

This results in a **False Negative Rate (FNR) of 15%**, indicating that 15% of high-risk patients were not identified. This example provides a reference point for understanding how metrics are calculated and interpreted within the system. Note: values in this example are illustrative only;

actual evaluation results for the pilot dataset are reported in the accompanying project report submitted to the Capstone professor.

4.4 From Outcomes to Metrics and Dashboard Insights

The outcome classifications described above are aggregated within the analytics environment to produce performance metrics at the population and subgroup level.

- Outcome counts (TP, FN, FP, TN) are grouped across patient populations
- Metrics such as FNR, recall, and PPV are calculated for each subgroup
- Results are passed to the dashboard layer for visualization

The dashboard presents these metrics in a structured format, enabling stakeholders to:

- Compare performance across subgroups
- Identify disparities in outcomes
- Monitor trends over time
- Detect threshold breaches

For example, while the overall FNR may be 15%, individual subgroups may experience higher or lower rates. These differences are highlighted in the dashboard and may trigger governance review.

4.5 Equity Dimensions

To ensure equitable performance, metrics are evaluated across key dimensions:

Dimension	Variables Included
Demographic	Age band, sex, race/ethnicity, Indigenous identity
Geographic	Health zones and urban/rural classification
Socioeconomic	SES quintile, insurance status
Access-related	Primary care access, EHR source, language

These dimensions are applied within the analytics environment and exposed as filters within the dashboard.

4.6 Metric Framework Overview

Metrics are organized into three tiers based on their role in monitoring and governance:

Tier	Purpose	Role
Tier 1	Critical safety and fairness	Drives governance decisions
Tier 2	Supporting monitoring metrics	Provides context and reliability checks
Drift Metrics	Temporal monitoring	Detects changes over time

This structure ensures that critical safety signals are prioritized while maintaining visibility into broader system behavior.

4.7 Tier 1 — Critical Safety Metrics

Using the example above:

- **FNR = 3 ÷ 20 = 15%**
- **Recall = 17 ÷ 20 = 85%**

These metrics are calculated within the analytics environment and displayed in the dashboard as key indicators of model performance.

Metric	What it Measures	Why it Matters	Threshold
False Negative Rate (FNR)	Missed diagnoses	Direct link to patient safety risk	Gap ≤ 8%
Recall	Detection of true cases	Complements FNR	≥ 85%
Positive Predictive Value (PPV)	Accuracy of positive predictions	Reduces unnecessary burden	Gap ≤ 12%
Calibration (ECE)	Accuracy of predicted probabilities	Supports decision-making	< 0.10

For example, if 25 patients are flagged as high risk and 17 are correct, the **PPV is 68%**.

These metrics are visualized using subgroup comparisons and threshold-based indicators within the dashboard. False Negative Rate (FNR) is prioritized as the primary metric due to its direct relationship to patient harm. In the context of pediatric diabetes, missed diagnoses may lead to diabetic ketoacidosis (DKA) within 24–48 hours, making systematic misses across certain populations both an equity issue and a patient safety risk.

4.8 Tier 2 — Monitoring Metrics

Metric reliability depends on the number of observations within each subgroup. If outcome counts are small, estimates may be unstable.

Metric	Purpose	Threshold
Sample Size	Ensures reliability	Suppress if n < 50; caution if 50–100
Missingness Rate	Tracks data completeness per row	Flag subgroup if >10% rows imputed

These controls are applied within the analytics environment and reflected in the dashboard to prevent misinterpretation.

4.9 Drift Detection Metrics

Over time, model behavior may change due to shifts in data or population characteristics.

For example:

- Initial FNR = 15%
- Later FNR = 20%

This increase represents **performance drift**.

Metric	What it Detects	Threshold
Population Stability Index (PSI)	Changes in input data distribution	> 0.2 triggers review
Performance Drift	Change in FNR or recall	+5% FNR
Missing Data Drift	Changes in data completeness	+10%
Calibration Drift (ECE)	Change in prediction reliability	+0.05 ECE from baseline; hard fail if any subgroup ECE > 0.15

How these metrics are interpreted:

- PSI identifies shifts in input data distributions, indicating potential changes in the patient population.
- Performance drift measures changes in key metrics compared to baseline.
- Missing data drift tracks changes in data completeness, which may affect model reliability.
- Calibration drift measures whether predicted probabilities remain aligned with actual outcomes.
- These metrics are computed in the analytics environment and visualized in the dashboard as trends over time.

4.10 Timing of Evaluation and Outcome Availability

Performance metrics rely on confirmed clinical outcomes and are therefore calculated retrospectively.

- Predictions are generated in real time
- Outcomes are recorded during follow-up care
- Metrics are calculated once sufficient outcome data is available

In contrast, drift metrics can be calculated earlier, providing **leading indicators** of potential issues before performance degradation is confirmed.

4.11 Link to Governance Decisions

Metrics displayed in the dashboard are evaluated within the governance layer to support decision-making.

Condition	Outcome	Action
Within thresholds	Pass	Continue monitoring
Minor deviations	Needs Review	Investigate/escalate to committee for approval/decision
Significant issues	Fail	Immediate action (Pause deployment - Pending remediation)

Governance outcomes (Pass, Needs Review, Fail) are determined using a standardized escalation framework based on predefined metric thresholds.

- **Pass**
  All Tier 1 metrics are within threshold, no calibration violations are present, and all subgroups meet minimum sample size requirements.
  Action: Continue routine monitoring.
- **Needs Review**
  Triggered when a metric exceeds its threshold by a small margin and may still be clinically acceptable with further review.
- **Fail**
  Triggered when performance, fairness, or calibration issues indicate unacceptable patient safety or equity risk.

For instance, the distinction between Needs Review and Fail is determined by a 20% exceedance rule. A metric that exceeds its threshold by less than 20% triggers a Needs Review outcome; a metric that exceeds its threshold by 20% or more, or where two or more metrics breach their thresholds simultaneously, triggers a Fail. For example, an FNR disparity of 9.5% (threshold ≤8%) represents a 19% exceedance and yields Needs Review, whereas a disparity of 9.7% (≥20% over threshold) yields Fail. In addition, any subgroup ECE exceeding 0.15 triggers an immediate Fail regardless of other metric results.

This graduated structure ensures that governance decisions are proportionate to risk, preventing both unnecessary deployment blocks for minor deviations and inadequate response to systematic failures.

4.12 Governance Considerations and Recommendations

To ensure the safe and equitable use of AI models, the evaluation framework applies the following principles:

- **Prioritize False Negative Rate (FNR)**
    - Focus on missed diagnoses due to their direct link to patient harm and safety risks
- **Assess disparities using absolute gaps**
    - Use absolute differences across subgroups to provide clearer, more clinically meaningful comparisons. Absolute gaps provide a more direct representation of differences in patient outcomes and are easier to interpret in a clinical context.
- **Include calibration as a core metric**
    - Ensure predicted risk scores align with actual outcomes to support reliable decision-making
- **Exclude metrics with limited clinical interpretability**
    - Avoid measures that do not directly inform patient safety or actionable insights

Our recommendations include:

- **Trigger governance review based on FNR disparities**
    - Initiate review when subgroup differences exceed **8 percentage points**
- **Monitor calibration alongside FNR**
    - Ensure both detection accuracy and risk reliability are maintained
- **Report metrics with subgroup sample sizes**
    - Provide context for interpretation and ensure transparency, particularly for smaller populations

5.0 AI Oversight, Monitoring, and Lifecycle Governance

Building on the Gatekeeper monitoring framework and the defined performance and fairness metrics outlined in earlier sections, effective oversight of clinical AI systems within Alberta Health Services (AHS) is operationalized through a multidisciplinary AI Oversight Committee. This structure integrates governance, continuous monitoring, and lifecycle management into a single accountable model, ensuring that AI systems—such as the pediatric diabetes diagnostic support use case—remain safe, effective, and equitable throughout real-world deployment.

This approach ensures that monitoring outputs generated by the Gatekeeper system are not only technically sound, but also interpreted within appropriate clinical, operational, and ethical contexts.

5.1 Committee Composition and Roles

The AI Oversight Committee is responsible for interpreting outputs from the Gatekeeper monitoring system and making governance decisions based on model performance, equity, and drift signals. It is composed of key stakeholder groups, each contributing distinct expertise to ensure balanced, transparent, and accountable decision-making:

- **Clinical Leadership (e.g., pediatric diabetes specialists)**

- Interpret model outputs in the context of patient care, particularly diagnostic accuracy and risks such as diabetic ketoacidosis (DKA)
    - Assess whether observed performance differences have meaningful clinical impact
- **Data and Analytics Teams**
    - Develop and operate the Gatekeeper monitoring pipelines
    - Generate and validate performance, subgroup equity, and drift metrics
    - Ensure outputs are statistically valid, reproducible, and appropriately interpreted
- **Digital Health and Clinical Informatics Leaders**
    - Ensure that model outputs and governance decisions align with clinical workflows and health information systems
    - Assess operational feasibility of any remediation actions (e.g., model updates or workflow changes)
- **Privacy and Legal Advisors**
    - Ensure compliance with privacy legislation, ethical standards, and data governance requirements
    - Provide oversight when sensitive attributes (e.g., Indigenous identity) are used in monitoring and evaluation
- **Quality and Patient Safety Representatives**
    - Evaluate risks related to missed diagnoses, model errors, and unintended consequences
    - Align governance decisions with patient safety priorities and quality improvement frameworks
- **Patient and Caregiver Representatives**
    - Provide perspectives on fairness, transparency, and trust
    - Ensure governance decisions reflect patient-centered values and lived experience

This multidisciplinary structure ensures that Gatekeeper outputs are interpreted holistically and translated into informed, accountable decisions.

5.2 Continuous Monitoring and Detection

At the core of the framework is the continuous evaluation of both overall model performance and subgroup equity, as operationalized through the Gatekeeper system. Monitoring outputs are translated into governance actions through predefined thresholds that define acceptable performance and equity boundaries. Escalation is triggered when overall performance declines beyond acceptable limits (performance drift) or subgroup disparities exceed defined thresholds (equity risk . When monitoring identifies potential issues, decisions are made through coordinated clinical, technical, and governance review. A risk-based approach is used to guide interventions:

- **Moderate-risk issues (Needs Review)**
    - May include minor performance declines or emerging subgroup disparities
    - Actions may include increased monitoring, threshold recalibration, or root cause analysis
- **High-risk issues (Fail)**

- May include significant performance degradation or major inequities (e.g., disproportionately high false negative rates for specific populations)
- Actions may include model retraining, data correction, workflow adjustments, or temporary suspension of the model

This structured approach ensures that responses are proportionate to risk and aligned with patient safety and clinical impact. Subsequent interventions inform model updates and data improvements. Learnings are fed back into model development and data governance processes

6.0 Governance Cadence and Oversight Rhythm

To support consistent and proactive oversight, governance activities follow a structured cadence:

- **Monthly Monitoring Reviews**
  - Routine performance and subgroup equity assessment
  - Early detection of drift or disparities
- **Quarterly Governance Reviews**
  - Trend analysis and threshold validation
  - Evaluation of governance effectiveness and policy alignment
- **Ad-hoc Escalation Reviews**
  - Triggered by high-risk events (e.g., Fail outcomes or critical equity concerns)
  - Enable rapid response to mitigate patient safety risks

This cadence balances proactive monitoring with the ability to respond quickly to emerging issues.

6.1 Transparency, Reporting, and Accountability

Effective communication of monitoring outcomes is essential to ensure clarity, accountability, and trust. Standardized governance reports can include performance metrics, subgroup comparisons, drift indicators, governance outcomes and recommended actions. Reporting is tailored to stakeholders (e.g. clinicians receive concise, actionable summaries; governance stakeholders receive detailed analyses; Where appropriate, plain-language summaries are provided to patients and caregivers to support transparency. All outputs contribute to a clear audit trail, ensuring that decisions are traceable, reproducible, and aligned with governance policies.

6.2 Recourse and Escalation Pathways

The framework includes structured pathways for identifying and resolving concerns arising from both system-generated signals and stakeholder observations. Concerns may originate from multiple sources:

- **Clinicians** observing unexpected or inconsistent model behavior in practice

- **Technical teams** identifying anomalies in model performance, data quality, or drift indicators
- **Governance reviewers** interpreting dashboard outputs and identifying threshold breaches
- **Patients or caregivers** raising concerns related to fairness, transparency, or trust

All concerns follow a standardized escalation process:

- **Issue identification and reporting**, triggered either by monitoring alerts (e.g., FNR disparities, drift signals) or stakeholder input
- **Technical investigation**, including validation of model outputs, data inputs, and system behavior
- **Clinical and governance review**, assessing potential patient safety, equity, and operational impact
- **Remediation and resolution**, which may include model recalibration, retraining, increased monitoring, or temporary suspension

This structured approach ensures that concerns identified through the Gatekeeper monitoring system are consistently evaluated and resolved in a transparent and accountable manner.

6.3 Auditability and Traceability

To support regulatory compliance, internal accountability, and continuous improvement, the framework incorporates comprehensive auditability and traceability mechanisms:

- All monitoring outputs, including performance metrics, drift indicators, and subgroup analyses, are systematically logged
- Governance decisions (e.g., Pass, Needs Review, Fail) and any overrides are documented and restricted to authorized personnel
- Each decision is fully traceable to:
  - Model version
  - Data snapshot used for evaluation
  - Applied thresholds and governance policies

This ensures that all decisions generated through the Gatekeeper system are reproducible, auditable, and defensible, supporting both compliance requirements and ongoing refinement of the monitoring framework.

7.0 Canadian Healthcare Examples Supporting AI Governance Oversight

SickKids:

At The Hospital for Sick Children, artificial intelligence governance is operationalized through SickKids Artificial Intelligence (SKAI), an enterprise-wide initiative that supports the responsible development, deployment, and scaling of AI systems across clinical, research, and operational domains. SKAI brings together clinicians, data scientists, digital health teams, and privacy and ethics stakeholders to ensure that AI models undergo rigorous pre-deployment validation, including:

- **Pre-deployment validation**, including clinical relevance and performance assessment
- **Ethical and privacy review**, aligned with institutional governance processes
- **Deployment support**, ensuring models are appropriately integrated into clinical workflows
- **Post-deployment evaluation**, including ongoing review of model behavior and performance in practice

Governance is embedded within existing institutional processes, with strong emphasis on patient safety, pediatric-specific considerations, and responsible data use. More information can be found here: https://www.sickkids.ca/en/about/sickkids-artificial-intelligence/

Rather than relying on a centralized, automated monitoring system, SickKids conducts post-deployment oversight through ongoing evaluation, clinician feedback, and periodic reassessment of model performance. This approach enables contextual and clinically informed interpretation of AI behavior but is typically implemented at the individual model level rather than through standardized, system-wide monitoring frameworks. For Alberta Health Services (AHS), this highlights the importance of multidisciplinary governance and pediatric clinical oversight, while also reinforcing the opportunity to extend these practices through the proposed Gatekeeper framework by introducing structured, continuous monitoring of performance, drift, and equity using defined metrics and thresholds.

CHEO

At Children's Hospital of Eastern Ontario, artificial intelligence is being integrated into pediatric care through both clinical innovation and research initiatives. CHEO publicly highlights several AI-enabled efforts, including predictive screening tools and machine learning models built on electronic health record (EHR) data, as described in its overview of AI initiatives (https://www.cheo.on.ca/en/about-us/artificial-intelligence-at-cheo.aspx). In addition, the CHEO Research Institute identifies artificial intelligence as a core research area supporting advanced analytics and clinical decision support (https://www.cheoresearch.ca/research/areas/artificial-intelligence/). A notable example is the ThinkRare algorithm, which uses EHR data to help identify children who may have undiagnosed rare genetic conditions (https://www.cheoresearch.ca/about-us/media/news/cheo-research-institutes-breakthrough-thinkrare-algorithm-launches-national-expansion/). These initiatives demonstrate that CHEO's approach to AI is grounded in clinical validation, research oversight, and multidisciplinary collaboration rather than a centralized, system-wide monitoring platform.

Post-deployment oversight at CHEO is typically conducted through research-led evaluation, clinician feedback, and iterative validation processes, rather than through standardized dashboards or automated monitoring frameworks for drift and subgroup bias. This reflects a governance model that prioritizes clinical safety, data governance, and contextual interpretation, particularly important in pediatric healthcare settings. For Alberta Health Services (AHS), CHEO reinforces the importance of embedding AI governance within clinical and research workflows, while also highlighting a broader gap across healthcare organizations: the absence of structured, continuous monitoring systems for evaluating model performance, equity, and drift at scale. The proposed Gatekeeper framework addresses this gap by introducing a system-level monitoring layer that enables consistent, transparent, and ongoing governance of AI models in real-world clinical environments.